

## SYNTEEN SF190 SOIL REINFORCEMENT GEOGRID

**SF190** is a uniaxial geogrid composed of high molecular weight, high tenacity multifilament polyester yarns that are woven into a stable network placed under tension and then coated with PVC to provide long-term durability. SF Geogrids are inert to biological degradation and are resistant to naturally encountered chemicals, alkalis and acids. Synteen SF series geogrids are used in a wide variety of soil reinforcement applications including segmental retaining walls (SRW), reinforced soil slopes (RSS), and mechanically stabilized earth (MSE) walls.

| MECHANICAL PROPERTIES                       | TEST METHOD | UNIT          | MINIMUM AVERAGE ROLL VALUE (MARV) <sup>1</sup> |
|---------------------------------------------|-------------|---------------|------------------------------------------------|
| Ultimate Tensile Strength (MD)              | ASTM D 6637 | lbs/ft (kN/m) | 20,560 (300.0)                                 |
| Creep Reduced Strength (MD) <sup>2</sup>    | ASTM D 5262 | lbs/ft (kN/m) | 13,615.9 (198.7)                               |
| Long Term Design Strength (MD) <sup>3</sup> | NCMA 97     | lbs/ft (kN/m) | 11,788.7 (172.0)                               |

| PHYSICAL PROPERTIES                              | TEST METHOD    | UNIT                              | ROLL VALUE                |
|--------------------------------------------------|----------------|-----------------------------------|---------------------------|
| Aperture Size                                    | Measured Value | in X in (mm X mm)                 | 0.51 X 1.00 (13.0 X 25.4) |
| Roll Dimensions <sup>4</sup><br>(Width x Length) | Measured Value | ft (m)                            | 12 X 150 ( 3.65 X 45.72 ) |
| Roll Area                                        | Measured Value | Yd <sup>2</sup> (m <sup>2</sup> ) | 200 ( 167 )               |

<sup>1</sup> Minimum Average Roll Values (MARV) are calculated as the typical minus two (2) standard deviations. Statistically, it yields a 97.7% degree of confidence that any sample taken from quality assurance testing will exceed the value reported.

<sup>2</sup> Creep Reduced Strength is based on a 75-year design life at 20 °C.  $RF_{CR} = 1.51$  is used based on NTPEP evaluation report.

<sup>3</sup> Long-term design strength (LTDS) is calculated for a 75-year design life at 20 °C, silty sand ( $D_{50} = 0.9\text{mm}$ ) backfill, and standard soil pH range 3 - 9.  $RF_{CR} = 1.51$ ;  $RF_{ID} = 1.05$ ;  $RF_D = 1.1$

<sup>4</sup> Custom roll widths and lengths are available upon request.

NTPEP evaluation report is available upon request.

LTDS and Reduction factors for the project specific requirements such as design life, soil type and soil pH beyond standard range 3-9 are available upon request.

**SYNTEEN** Technical Fabrics reserves right to amend product specifications without notice. Users / Buyers shall verify the product compliance with the project specifications or design requirements.